A0/A1 fine-aim sensor: slit design

MQ Sentinel · ENGG2000 V3 · Option 1 (slit split-detector) · Draft A · 2026-10-25 · Prepared by Y. Cui · units mm

Draft for review and cardboard testing. All dimensions and readings are calculated from geometry (TSSP58P38 datasheet: body 5.0 × 6.95 × 4.8, lens dome about 4 mm wide). Nothing here has been bench-tested.

1. How it works

Both TSSP58P38 receivers face straight ahead, side by side, inside a closed black box. The only opening is a 4 × 10 mm vertical slit, 20 mm in front of the lenses, centred on the gap between them. Beacon light through the slit forms a 4 mm strip on the sensors. When the beacon moves by an angle ϕ, the strip slides sideways by 20 × tan ϕ (0.35 mm per degree). At centre the strip is split evenly between A0 and A1. Off centre, one sensor gets more of it, and e = (L − R) / (L + R) shows how far off you are and which way.

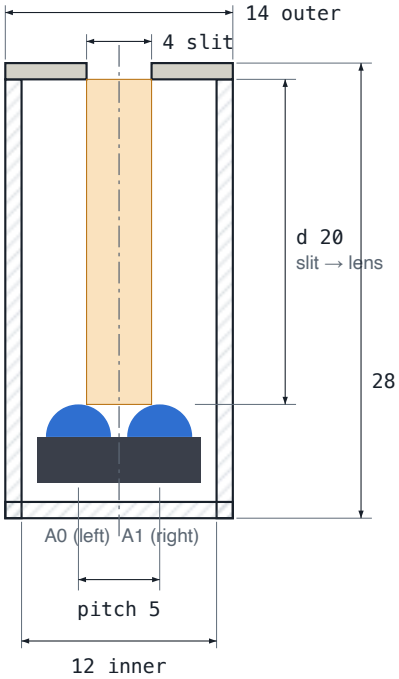
Direction is reversed compared with the T design. Like a pinhole camera, light from a beacon on the **left** lands on the **right** sensor (A1). So with this design, A1 > A0 means the beacon is to the LEFT. The LEFT/RIGHT labels in `a1a2_test` must be swapped (or the sign of e flipped) before using it with the slit.

2. Key dimensions

Feature	Value	Why
Slit width	`4 (spares 3, 5)`	Sets the fine-aim zone and the slope of e
Slit height	`10`	Lens fully lit vertically within ±6.4°; blocks floor and ceiling reflections
Slit to lens front face (d)	`20`	Strip moves 0.35 mm per degree; fits inside the pipe
Sensor pitch	`5 (bodies touching)`	Lenses about 1 mm apart
Slit position	`centred on lens gap`	The most critical dimension: 0.2 mm off centre shifts the aim by about 0.6°
Box inner width × height	`12 × 10`	Holds both sensors with about 1 mm play
Box outer (1 mm walls)	`14 × 12 × 28`	W × H × depth, back wall to front face
Front plate thickness	`≤ 1`	A thick plate blurs the slit edges
Finish	`matte black`	Every face, inside and out; the slit is the only opening

3. Drawing (draft, 9:1)

TOP SECTION, AT LENS HEIGHT

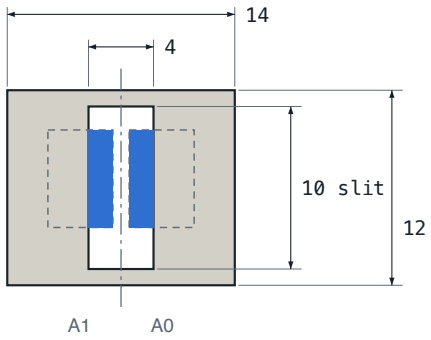


Beacon is toward the top of this view.
Hatched: box walls (cut). Grey: front plate.
Orange: light strip through the slit, beacon at 0°.
Dash-dot: centre line = laser direction.

Beacon moves φ → strip slides 20·tan φ the OTHER way.
Beacon 2.3° left → strip 0.8 mm right → A1 brighter.

FROM THE BEACON

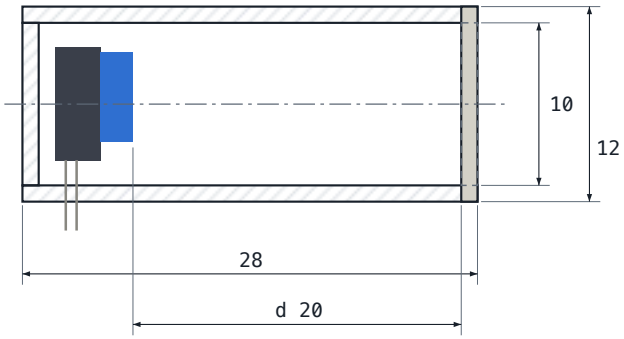
Seen from the target: A1 appears on the viewer's left.



Lens gap (about 1 mm) must sit in the middle of the slit.

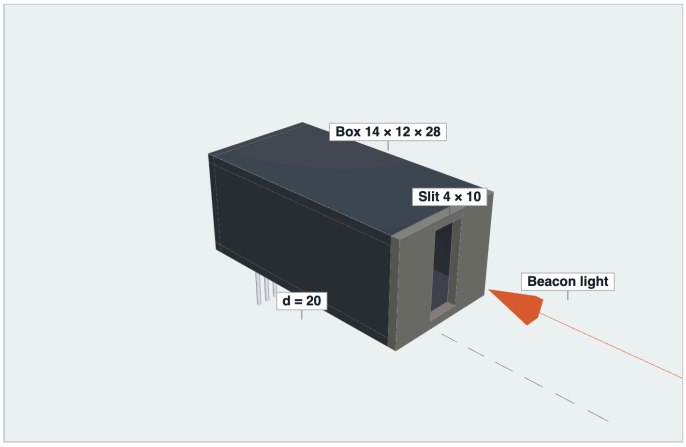
SIDE SECTION, THROUGH A0

Beacon is to the right.

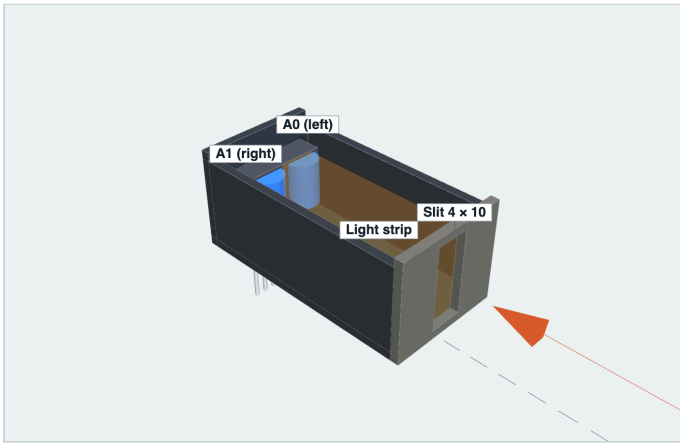


Dashed: slit outline in the front plate. Leads exit down through the floor.

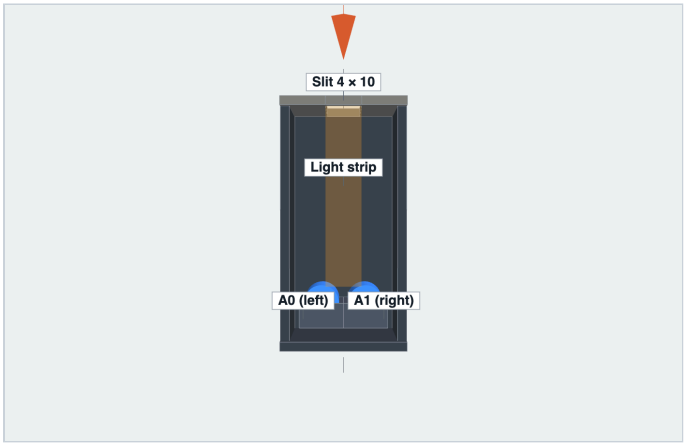
4. 3D views



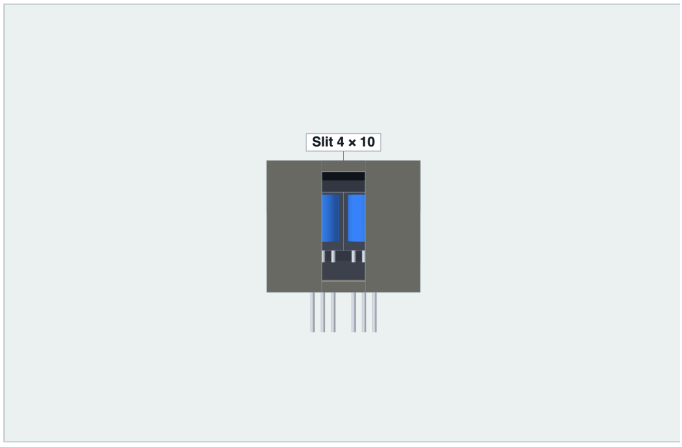
3/4 view, closed



3/4 view, roof removed



Top view, roof removed (beacon at top)



From the beacon (A1 on the viewer's left)

5. Expected readings (geometry only)

Share of the 4 mm light strip landing on each lens, slit width 4, d = 20, lens gap 1. Beacon to the **left**; to the right is the mirror image.

Beacon left by	At 1 m	Strip on A0 (mm)	Strip on A1 (mm)	e = (L−R)/(L+R)
0°	0 cm	1.50	1.50	0 (= E0 after calibration)
1°	1.7 cm	1.15	1.85	−0.23
2.3° (bullseye edge)	4 cm	0.70	2.30	−0.53
4°	7 cm	0.10	2.90	−0.93
4.3° and beyond	7.5 cm	0	all	−1: direction only
about 18°	32 cm	0	0	beacon lost

Slit width	Fine-aim zone (both see it)	e at 2.3°	Light let in
3	±2.9°	about −0.8	least; zone barely wider than the bullseye
4 (start)	±4.3°	about −0.53	medium
5	±5.7°	about −0.4	most; easiest to capture

The TSSP's pulse width is roughly logarithmic in light, so real changes in e will be smaller than these shares suggest. The bench sweep sets the real E0, slope and tolerance.

6. Cardboard build

1. Cut: floor and roof 14 × 28; two side walls 26 × 10; back wall 12 × 10; front plate 14 × 12 (about 1 mm card). Colour every piece matte black.
2. On the floor, draw the centre line. Mark the lens line 20 mm behind where the front plate will go.
3. Place both sensors facing straight ahead, bodies touching, the gap between them on the centre line and their lens fronts on the lens line. Push the leads through the floor and tape them underneath.
4. Glue the side walls (12 mm apart inside) and the back wall.
5. Cut the 4 × 10 slit in the front plate, centred left-right and top-bottom, with clean straight edges. Glue the plate on so the slit sits over the lens gap. Look through the slit: equal amounts of each lens should show either side of the gap.
6. Tape the roof on, so the front plate can be swapped for a 3 mm or 5 mm slit.

7. Bench test

1. Wiring as before: A0 OUT → pin A0, A1 OUT → pin A1, both VS → D4, GND; IR LED on D3 with a 100–330 Ω resistor.
2. Upload `a1a2_test`. Start with **no paper** over the LED, at 0.30 m, LED at lens height, aimed at the slit.
3. No LED: both read about 0 (NO SIGNAL). With LED at 0°: both read thousands to tens of thousands, no SAT, e near 0.
4. Sweep 0, ±1, ±2, ±3, ±4, ±6, ±10, ±15, ±20° by turning the box about the slit; record about 10 lines per angle.
5. Check: A1 should rise when the beacon is to the left. If it is A0 instead, the slit or sensors are swapped.